Sources

Folate naturally occurs in a wide variety of foods, including dark green leaf vegetables, fruits, nuts, soybeans, chickpeas, dairy products, poultry and meat, eggs, seafood, grains. Vegetables such as avocado, beetroot, spinach, liver, yeast, asparagus, kale, and brussels and sprouts are among the foods with the highest levels of folate.

Function

- Folic acid is essential for cell division and growth
- It is important for preventing birth defects.
- It is used in the synthesis of amino acid methionine.
- Tetrahydrofolate (THF) is needed to transfer one carbon units in the biosynthetic reactions.

Absorption and storage

Folates are absorbed in the small intestine. The body can store a small amount in liver. The excess is removed from liver via bile.

Deficiency

Folate deficiency signs include anemia (megaloblastic anemia) and dysfunction of the gastrointestinal tract. The anemia is caused by the abnormal blood cell division resulting in fewer and larger red blood cells.

Lack of folate in the diet can cause neural tube defects in an embryo of a pregnant woman. These can cause fatal birth defects.

viii. Pantothenic acid (Vitamin B5):

Pantothenic acid is found in every living cell including plant, animal and microbes. It is a part of coenzyme A which is an essential coenzyme involved in the catabolism of carbohydrates, proteins and fat.

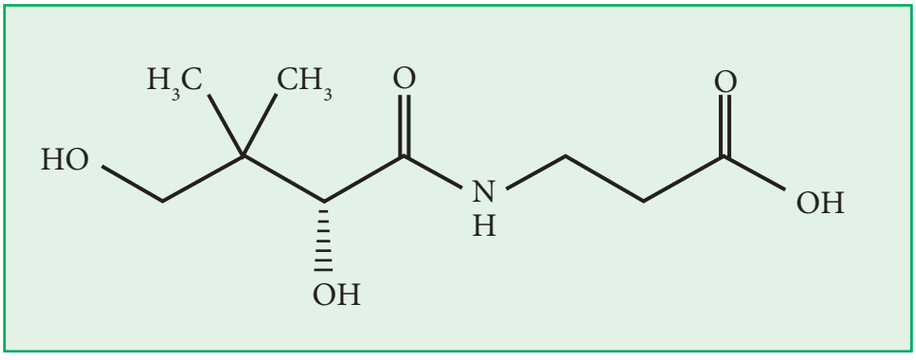

Fig. 8.28 Structure of pantothenic acid

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